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## **ME-7005(1)-CBGS**

### **B.E. VII Semester**

Examination, June 2020

## **Choice Based Grading System (CBGS)**

### **Robotics**

**Time : Three Hours**

**Maximum Marks : 70**

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. What is a Robot? Discuss various types of Robots. What is anatomy of a Robot?
2. Discuss various parts of Robots and their functions? What is pitch, yaw, roll, joint and payload as connected to Robots?
3. Compare, Pneumatic, Hydraulic, electric and mechanical drives as connected to Robots and discuss their individual applications.
4. Discuss following Grippers:  
Mechanical, Pneumatic and Hydraulic grippers, Magnetic grippers, Vacuum grippers, Two and three fingered grippers and external grippers.
5. Discuss following sensors and their applications in Robotics.
  - a) Position sensors
  - b) Range sensors
  - c) Proximity sensors
  - d) Touch sensors
  - e) Slip sensors

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6. Discuss concept of forward kinematics and reverse kinematics of manipulators with two and three degree of freedom.
7. Explain economic analysis of Robots. Discuss pay back method, EVAC method and Rate of Return Method.
8. Write short notes on followings:
  - a) RGV and AGV
  - b) Robot Programming
  - c) Gripper selection and design consideration
  - d) Need of Robots - unusual applications

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